

Creation Date 18-Oct-2010

Revision Date 22-Sep-2023

Revision Number 7

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                        |
|---------------------------|----------------------------------------|
| Product Description:      | <b>Cobalt, powder</b>                  |
| Cat No. :                 | <b>192850000; 192850050; 192850250</b> |
| Index No                  | 027-001-00-9                           |
| CAS No                    | 7440-48-4                              |
| EC No                     | 231-158-0                              |
| Molecular Formula         | Co                                     |
| REACH registration number | -                                      |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

#### EU entity/business name

Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

#### UK entity/business name

Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

#### Swiss distributor - Fisher Scientific AG

Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

#### E-mail address

begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:

Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

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## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

Acute oral toxicity  
Respiratory Sensitization  
Skin Sensitization  
Germ Cell Mutagenicity  
Reproductive Toxicity

Category 4 (H302)  
Category 1 Sub-category 1B (H334)  
Category 1 (H317)  
Category 2 (H341)  
Category 1B (H360F)

##### Environmental hazards

Chronic aquatic toxicity

Category 4 (H413)

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

**Danger**

#### **Hazard Statements**

H302 - Harmful if swallowed  
H317 - May cause an allergic skin reaction  
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
H341 - Suspected of causing genetic defects  
H350i - May cause cancer by inhalation  
H360F - May damage fertility  
H413 - May cause long lasting harmful effects to aquatic life  
May form combustible dust concentrations in air

#### **Precautionary Statements**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

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## Additional EU labelling

Restricted to professional users

### 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

May form explosible dust-air mixture if dispersed

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component      | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                         |
|----------------|-----------|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cobalt, powder | 7440-48-4 | EEC No. 231-158-0 | <=100    | Acute Tox. 4 (H302)<br>Resp. Sens. 1B (H334)<br>Skin Sens. 1 (H317)<br>Muta. 2 (H341)<br>Repr. 1B (H360F)<br>Carc. 1B (H350i)<br>Aquatic Chronic 4 (H413) |

REACH registration number

-

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| Ingestion                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| Self-Protection of the First Aider | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                 |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness,

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lightheadedness, chest pain, muscle pain or flushing

## **4.3. Indication of any immediate medical attention and special treatment needed**

Notes to Physician

Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **5.2. Special hazards arising from the substance or mixture**

Flammable. Containers may explode when heated. Dust can form an explosive mixture with air. Do not allow run-off from fire-fighting to enter drains or water courses. Fine dust dispersed in air may ignite.

#### **Hazardous Combustion Products**

Cobalt oxides.

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### **6.2. Environmental precautions**

Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

### **6.3. Methods and material for containment and cleaning up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical

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assistance.

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1D

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component      | European Union | The United Kingdom                                                                   | France | Belgium                               | Spain                                             |
|----------------|----------------|--------------------------------------------------------------------------------------|--------|---------------------------------------|---------------------------------------------------|
| Cobalt, powder |                | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Resp. Sens. |        | TWA: 0.02 mg/m <sup>3</sup> 8<br>uren | TWA / VLA-ED: 0.02<br>mg/m <sup>3</sup> (8 horas) |

| Component      | Italy | Germany | Portugal                               | The Netherlands                       | Finland                                   |
|----------------|-------|---------|----------------------------------------|---------------------------------------|-------------------------------------------|
| Cobalt, powder |       | Haut    | TWA: 0.02 mg/m <sup>3</sup> 8<br>horas | TWA: 0.02 mg/m <sup>3</sup> 8<br>uren | TWA: 0.02 mg/m <sup>3</sup> 8<br>tunteina |

| Component      | Austria                                                                                                                                                                  | Denmark                                                                               | Switzerland                                           | Poland                                     | Norway                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Cobalt, powder | TRK-KZGW: 2 mg/m <sup>3</sup><br>15 Minuten<br>TRK-KZGW: 0.4 mg/m <sup>3</sup><br>15 Minuten<br>Haut<br>TRK-TMW: 0.5 mg/m <sup>3</sup><br>TRK-TMW: 0.1 mg/m <sup>3</sup> | TWA: 0.01 mg/m <sup>3</sup> 8<br>timer<br>STEL: 0.02 mg/m <sup>3</sup> 15<br>minutter | Haut/Peau<br>TWA: 0.05 mg/m <sup>3</sup> 8<br>Stunden | TWA: 0.02 mg/m <sup>3</sup> 8<br>godzinach | TWA: 0.02 mg/m <sup>3</sup> 8<br>timer<br>STEL: 0.06 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated fume |

| Component      | Bulgaria                   | Croatia                                     | Ireland                                                                 | Cyprus | Czech Republic                                                                                                |
|----------------|----------------------------|---------------------------------------------|-------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------|
| Cobalt, powder | TWA: 0.1 mg/m <sup>3</sup> | TWA-GVI: 0.1 mg/m <sup>3</sup> 8<br>satima. | TWA: 0.02 mg/m <sup>3</sup> 8 hr.<br>STEL: 0.3 mg/m <sup>3</sup> 15 min |        | TWA: 0.05 mg/m <sup>3</sup> 8<br>hodinách. inhalable<br>fraction of aerosol<br>Ceiling: 0.1 mg/m <sup>3</sup> |

| Component      | Estonia                                    | Gibraltar | Greece                     | Hungary                                     | Iceland                                                                                                              |
|----------------|--------------------------------------------|-----------|----------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Cobalt, powder | TWA: 0.05 mg/m <sup>3</sup> 8<br>tundides. |           | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> 8<br>óraban. AK | TWA: 0.02 mg/m <sup>3</sup> 8<br>klukkustundum. dust<br>and fume<br>Ceiling: 0.04 mg/m <sup>3</sup><br>dust and fume |

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| Component      | Latvia                     | Lithuania                        | Luxembourg | Malta | Romania                                                                    |
|----------------|----------------------------|----------------------------------|------------|-------|----------------------------------------------------------------------------|
| Cobalt, powder | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> IPRD |            |       | TWA: 0.05 mg/m <sup>3</sup> 8 ore<br>STEL: 0.1 mg/m <sup>3</sup> 15 minute |

| Component      | Russia                                                                           | Slovak Republic             | Slovenia | Sweden                                        | Turkey |
|----------------|----------------------------------------------------------------------------------|-----------------------------|----------|-----------------------------------------------|--------|
| Cobalt, powder | TWA: 0.01 mg/m <sup>3</sup> 1108<br>Skin notation<br>MAC: 0.05 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> |          | TLV: 0.02 mg/m <sup>3</sup> 8 timmar. NGV Hud |        |

## Biological limit values

List source(s):

| Component      | European Union | United Kingdom | France                                                                                                               | Spain                                                                         | Germany |
|----------------|----------------|----------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------|
| Cobalt, powder |                |                | Cobalt: 0.001 mg/L blood end of shift at end of workweek<br>Cobalt: 0.015 mg/L urine end of shift at end of workweek | Cobalt: 15 µg/L urine end of workweek<br>Cobalt: 1 µg/L blood end of workweek |         |

| Component      | Italy | Finland                                                                                         | Denmark | Bulgaria | Romania                                                                         |
|----------------|-------|-------------------------------------------------------------------------------------------------|---------|----------|---------------------------------------------------------------------------------|
| Cobalt, powder |       | Cobalt: 130 nmol/L urine after the work phase or shift after a working week or exposure period. |         |          | Cobalt: 15 µg/L urine end of work week<br>Cobalt: 1 µg/L blood end of work week |

| Component      | Gibraltar | Latvia | Slovak Republic                    | Luxembourg | Turkey |
|----------------|-----------|--------|------------------------------------|------------|--------|
| Cobalt, powder |           |        | Cobalt: 30 µg/L urine not critical |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

MDHS30/2 Cobalt and cobalt compounds in air Laboratory method using flame atomic absorption spectrometry

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                             | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|---------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Cobalt, powder<br>7440-48-4 ( <=100 ) |                                  |                                     | DNEL = 40µg/m <sup>3</sup>         |                                       |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                             | Fresh water     | Fresh water sediment         | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|---------------------------------------|-----------------|------------------------------|--------------------|------------------------------------|--------------------------|
| Cobalt, powder<br>7440-48-4 ( <=100 ) | PNEC = 0.62µg/L | PNEC = 53.8mg/kg sediment dw |                    | PNEC = 0.37mg/L                    | PNEC = 10.9mg/kg soil dw |

| Component      | Marine water    | Marine water sediment | Marine water Intermittent | Food chain | Air |
|----------------|-----------------|-----------------------|---------------------------|------------|-----|
| Cobalt, powder | PNEC = 2.36µg/L | PNEC = 69.8mg/kg      |                           |            |     |

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|                     |  |             |  |  |  |
|---------------------|--|-------------|--|--|--|
| 7440-48-4 ( <=100 ) |  | sediment dw |  |  |  |
|---------------------|--|-------------|--|--|--|

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

**Skin and body protection** Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                          |                          |            |
|--------------------------|--------------------------|------------|
| Physical State           | Solid                    |            |
| Appearance               | Grey                     |            |
| Odor                     | Odorless                 |            |
| Odor Threshold           | No data available        |            |
| Melting Point/Range      | 1495 °C / 2723 °F        |            |
| Softening Point          | No data available        |            |
| Boiling Point/Range      | 2870 °C / 5198 °F        | @ 760 mmHg |
| Flammability (liquid)    | Not applicable           | Solid      |
| Flammability (solid,gas) | No information available |            |
| Explosion Limits         | No data available        |            |

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|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Flash Point                             | Not applicable           | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | No information available |                                   |
| Viscosity                               | Not applicable           | Solid                             |
| Water Solubility                        | Insoluble                |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Cobalt, powder                          | 5                        |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Particle characteristics                | No data available        |                                   |

## 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | Co                     |
| Molecular Weight  | 58.93                  |
| Evaporation Rate  | Not applicable - Solid |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

Strong oxidizing agents. Strong acids.

### 10.6. Hazardous decomposition products

Cobalt oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met



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| Component      | LD50 Oral                 | LD50 Dermal | LC50 Inhalation              |
|----------------|---------------------------|-------------|------------------------------|
| Cobalt, powder | LD50 = 6171 mg/kg ( Rat ) | -           | LC50 < 0.05 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

(d) respiratory or skin sensitization;

Respiratory

Skin

Sub Category 1B

Category 1

No information available

(e) germ cell mutagenicity;

Category 2

In vitro tests have shown mutagenic effects

(f) carcinogenicity;

Based on available data, the classification criteria are not met

This product contains one or more substances which are classified by IARC as carcinogenic to humans (Group I), probably carcinogenic to humans (Group 2A) or possibly carcinogenic to humans (Group 2B) The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component      | EU           | UK | Germany | IARC     |
|----------------|--------------|----|---------|----------|
| Cobalt, powder | Carc Cat. 1B |    | Cat. 2  | Group 2A |

(g) reproductive toxicity; Category 1B

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs

None known.

(j) aspiration hazard;

Not applicable  
Solid

Symptoms / effects, both acute and delayed

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## 11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

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| Component      | Freshwater Fish                                  | Water Flea | Freshwater Algae |
|----------------|--------------------------------------------------|------------|------------------|
| Cobalt, powder | LC50: > 100 mg/L, 96h static (Brachydanio rerio) |            |                  |

**12.2. Persistence and degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
**Persistence** May persist.  
**Degradability** Not relevant for inorganic substances.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**12.3. Bioaccumulative potential** Product has a high potential to bioconcentrate

| Component      | log Pow | Bioconcentration factor (BCF) |
|----------------|---------|-------------------------------|
| Cobalt, powder | 5       | No data available             |

**12.4. Mobility in soil** Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility. Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles

**12.5. Results of PBT and vPvB assessment** In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

**12.6. Endocrine disrupting properties**  
**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**12.7. Other adverse effects**  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)** According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information** Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

**Switzerland - Waste Ordinance** Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600

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<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### ADR

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### IATA

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

#### 14.5. Environmental hazards

No hazards identified

#### 14.6. Special precautions for user

No special precautions required.

#### 14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component      | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|----------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Cobalt, powder | 7440-48-4 | 231-158-0 | -      | -   | X     | X    | KE-06060 | X    | -    |

| Component      | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|----------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Cobalt, powder | 7440-48-4 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component | CAS No | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|--------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|           |        |                                                                     |                                                                               |                                                                                                       |

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|                |           |   |                                                                                                                                                                                                                |   |
|----------------|-----------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Cobalt, powder | 7440-48-4 | - | Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | - |
|----------------|-----------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component      | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Cobalt, powder | 7440-48-4 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component      | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                                                                                                   |
|----------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Cobalt, powder | WGK 3                                 | Class II : 0.5 mg/m <sup>3</sup> (Massenkonzentration)<br>Krebserzeugende Stoffe - Class I : 0.05 mg/m <sup>3</sup> (Massenkonzentration) |

| Component      | France - INRS (Tables of occupational diseases)                              |
|----------------|------------------------------------------------------------------------------|
| Cobalt, powder | Tableaux des maladies professionnelles (TMP) - RG 65,RG 70,RG 70bis,RG 70ter |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

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## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H317 - May cause an allergic skin reaction  
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
H341 - Suspected of causing genetic defects  
H350 - May cause cancer  
H350i - May cause cancer by inhalation  
H360F - May damage fertility  
H413 - May cause long lasting harmful effects to aquatic life

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Training Advice

Chemical incident response training.

**Creation Date** 18-Oct-2010

**Revision Date** 22-Sep-2023

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information

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relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**